Data Science Project

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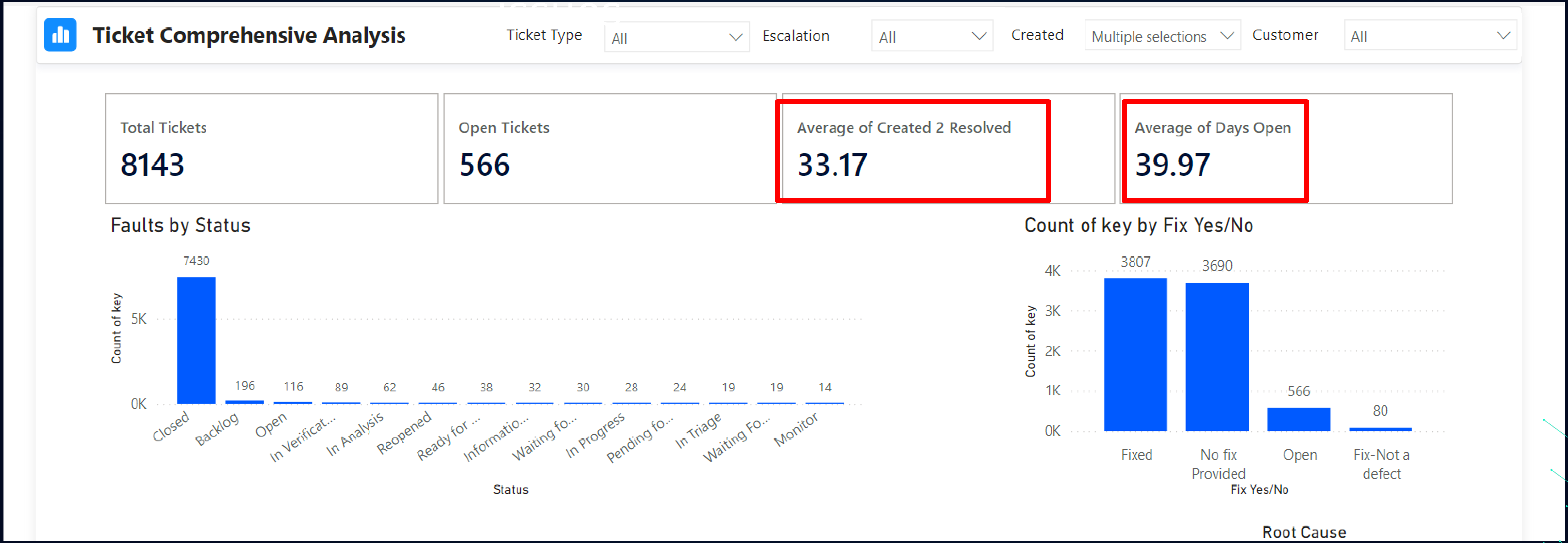
Research Question

Can obstacles in the **fault resolution** process be predicted?

Overview

The Problem

Long resolution times for



Research Question

Business Problems

01 Excessive developer time on issues

02 Workflow disruptions

03 Issues reaching customers

04 Limited scalability across departments

Data Collection

Changelog

Dataset

Key Data Elements:

- **ID:** Unique identifier for each change record.
- **Issue_Key:** Unique identifier for the ticket.
- **FIELD:** The attribute changed (e.g., status, team).
- **OLDSTRING / NEWSTRING:** The previous and updated values of the field.
- **CREATED:** Timestamp of the change.

Data Collection

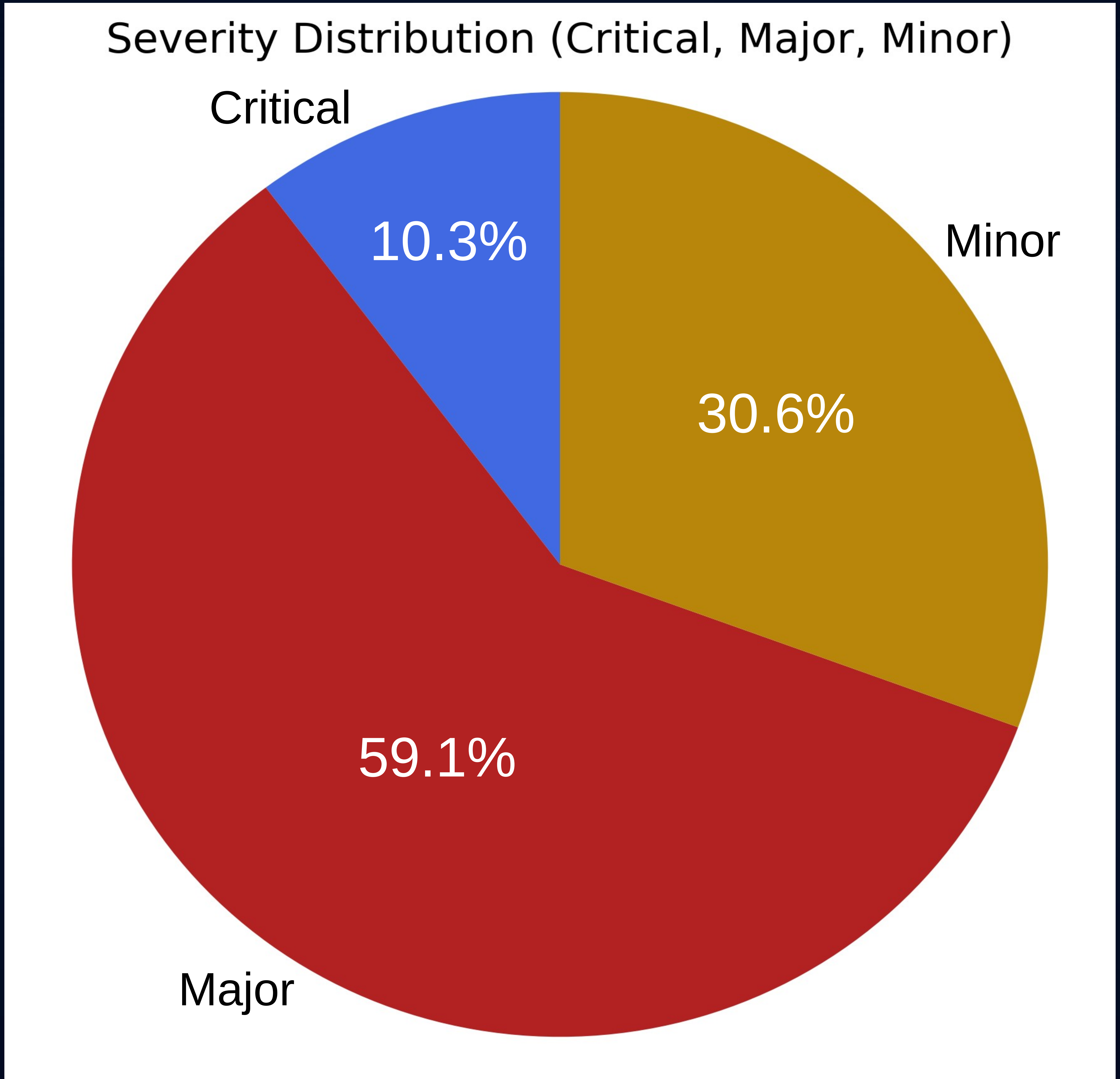
Changelog

Dataset

Insights & Use Cases:

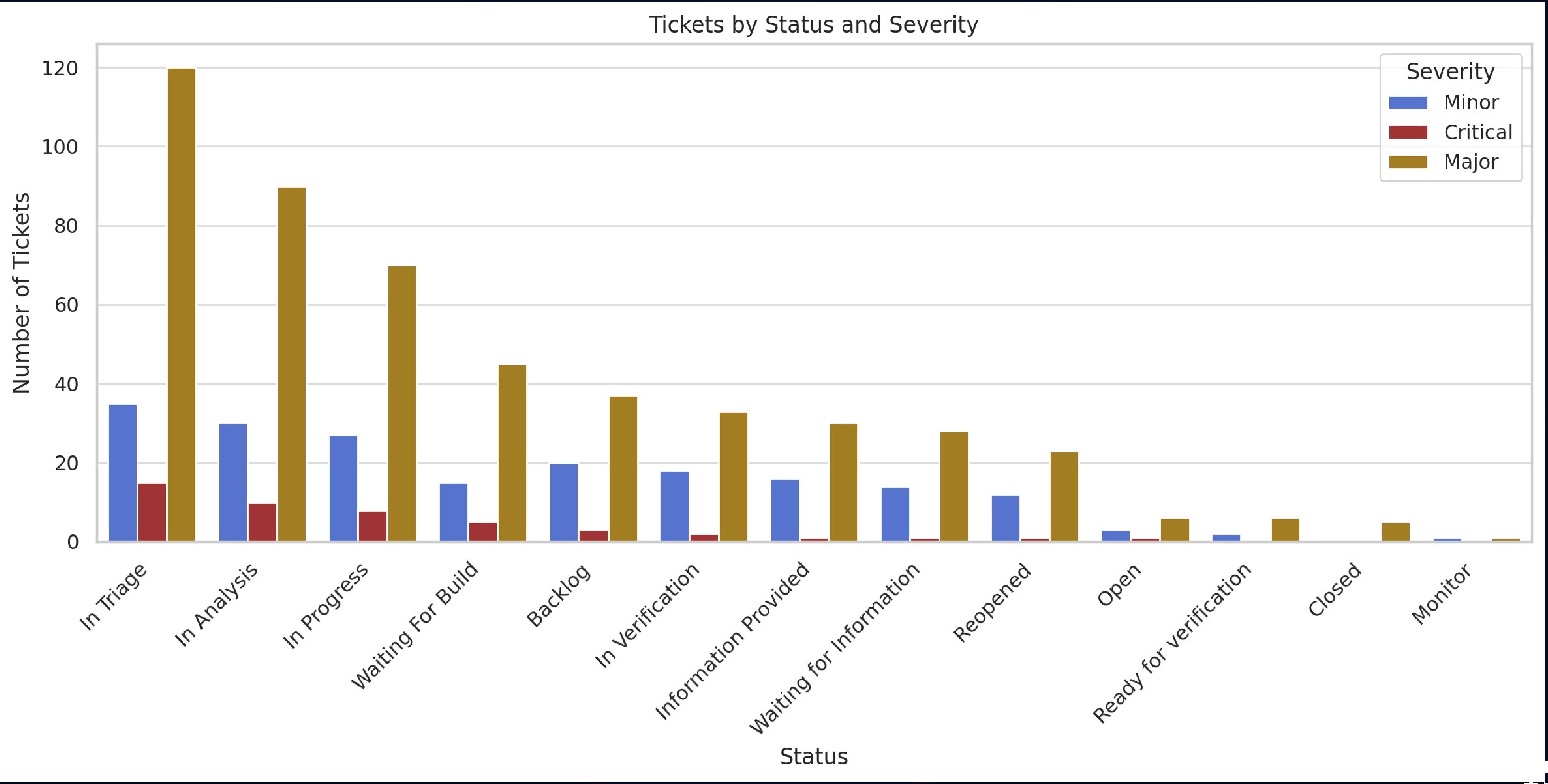
- **Track Progress: Monitor status transitions like "Open" → "In Progress.“**
- **Optimize Workflows: Identify bottlenecks in ticket handling.**
- **Team Analysis: Highlight frequent team changes for process improvement.**
- **Severity Prioritization: Focus on Critical and Major issues.**
- **Time in Status: Measure delays for efficiency**

Fault Types

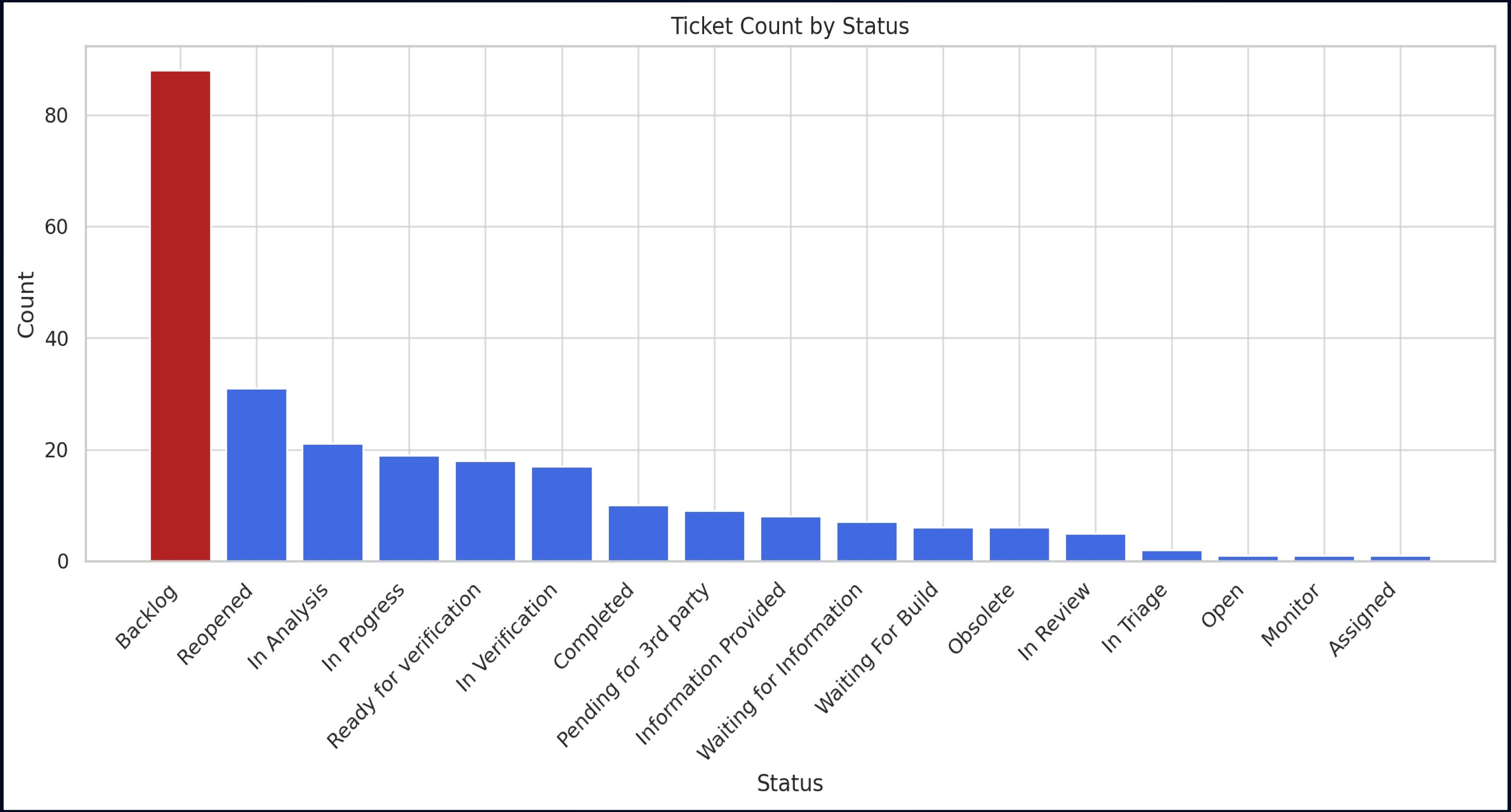

Severity Distribution (Critical, Major, Minor)
Critical
10.3%
Minor
30.6%
59.1%
Major

Number of Times Each Status Was Assigned

Number of Times Each Status Was Assigned



Current Status Distribution



Overview

The process

1. Raw Data (Tickets)
2. Eembedding via GPT-3
3. Neural Network
4. Predicted team name

The Model

Why a Neural Network?

- **Our task is a classification problem involving both structured and unstructured data.**
- **Neural networks are effective in handling mixed data types and capturing complex non-linear patterns.**
- **The use of text embeddings from ChatGPT-4 required a model that can integrate high-dimensional vector inputs with categorical features.**

The Model

Model Structure

- Feedforward neural network
- One hidden layer with ReLU activation
- Output layer matches number of support teams

Hyperparameter Tuning - using Optuna

- Parameters: hidden_dim, learning_rate, batch_size, weight_decay, patience.
- Evaluation via 4-fold cross-validation for robust performance

The Model

The Model

Training the model

- Trained a fresh model on each fold
- Optimizer: Adam
- Loss: Cross-Entropy
- Early stopping used to prevent overfitting

Metrics Tracked via W&B

- Loss & Accuracy (Train & Validation)
- Precision, Recall, F1-score (macro)
- AUC-ROC (multi-class, ovo

The Model

Model Selection

- Chose the model with lowest average validation loss
- Saved best model weights for future use

Ethical Dilemmas

🔐 **Use of GPT-3 and Privacy Concerns:**
Ensure anonymization before sending data to the model.

📊 **Model Transparency:**
The system should explain how predictions are made.

⚠ **Bias in Historical Data:**
Training data may reinforce existing biases.

👥 **Impact on Teams:**
Predictions may affect workload distribution and perceived fairness

Thank you!